POLYNEXUS HOSPITAL MANAGEMENT SYSTEM

Delivering to a Client

Build on your PC, ship one bundle, licence each hospital

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THE JOURNEY AT A GLANCE

Part	What happens	Who	Where
1. Build	Make the install bundle	You	Your PC (WSL)
2. Deliver	Send the bundle to the hospital	You	USB drive or link
3. Install	First run of the installer	Hospital IT	Client's server
4. Licence	Request and approve the licence	Licence team + SaaS Owner	SaaS console
5. Activate	Second run of the installer	Hospital IT	Client's server
6. Domain	Put it on hms.hospital.com with HTTPS	Hospital IT	Client's server
7. Backups	Send nightly backups off the server	Hospital IT	Client's server
8. Later	Updates and renewals	Both	

Before you start

You never copy code to the client. Everything is built on your PC and only the finished bundle goes to the hospital. The bundle holds the compiled program (no source code) and the installers. The client's server needs only **Docker** — no Python, Node or source code — and can run **Linux** or **Windows**.

One bundle serves every client. What differs per hospital is only its licence file: features, number of users, expiry date, and the machine it's bound to.

Never send: the `Backend` or `Frontend` folders, `venv`, any `.env` file, or your private signing key (`license_signing_key.pem`).

The private signing key lives only on the SaaS server, with one offline backup in a vault. Licences are created only in the SaaS console, never on a PC.

1 Build the bundle

On your PC, with **Docker Desktop running**, open **WSL (Ubuntu)**:

```
# WSL (Ubuntu) on your PC
cd /mnt/e/Aniket/next/Hospital/Hospital/Frontend
git pull
git lfs pull
cd ../Backend
git pull
make bundle VERSION=1.0.1
```

Before `make bundle`**:** in the SaaS console, On-Premise Licences → **Revocation list**, save it as `Backend\apps\licensing\revocations.lic`. Every release carries the list, so a licence you revoked stops working on a server once it upgrades. `make bundle` warns if the file is missing.

- The bundle contains **both** the backend and the frontend (with nginx), plus the PostgreSQL and Redis images.
- `git lfs pull` downloads the 3D anatomy model (168 MB), which is built into the frontend. Without it the build stops and tells you to run it.
- Use a new, higher version number for every release (1.0.1, 1.0.2, 1.1.0, ...).
- The first build takes 15–30 minutes; later builds are faster.
- Result: `Backend\dist\bundle-1.0.1.tar.gz` (about 530 MB).

2 Deliver the bundle

Send `bundle-1.0.1.tar.gz` to the hospital: USB drive, Google Drive or OneDrive link, or secure file transfer. It's safe to share: it contains no source code and no secrets.

So the hospital can confirm the file arrived complete, send its checksum with it:

```
# WSL (Ubuntu) on your PC
sha256sum dist/bundle-1.0.1.tar.gz
```

The hospital compares it with `Get-FileHash bundle-1.0.1.tar.gz` (Windows) or `sha256sum bundle-1.0.1.tar.gz` (Linux).

3 Install on the client's server

Hospital IT runs this from the folder where they saved the bundle.

Linux server (as root):

```
# Linux server
sudo mkdir -p /opt/hms
sudo tar -xzf bundle-1.0.1.tar.gz -C /opt/hms --strip-components=1
cd /opt/hms
sudo ./install.sh
```

Windows server (PowerShell as administrator; Docker Desktop with Linux containers):

```
# Windows server - PowerShell (administrator)
mkdir -Force C:\hms
tar -xzf bundle-1.0.1.tar.gz -C C:\hms --strip-components=1
cd C:\hms
powershell -ExecutionPolicy Bypass -File .\install.ps1
```

The installer asks for the **hostname or IP** staff will use, generates secrets, loads the program (no internet needed), and then prints a **Deployment ID** and a **Machine fingerprint**. It saves both to `config/REQUEST-LICENCE.txt`, then stops.

The hospital sends you `REQUEST-LICENCE.txt`.

> **Always install with `install.sh` / `install.ps1`, never with `docker compose up`.** The installer creates `.env` and loads the program first; without it, Docker stops with "*no .env here - run install.sh (Linux) or install.ps1 (Windows) first*" (older bundles: "*HMS_VERSION is not set in .env*"). That message means only that the installer hasn't been run. Nothing is broken. On Windows, PowerShell must be opened **as administrator** (not Command Prompt). If the bundle was unpacked by right-click → *Extract All*, the files end up in `bundle-1.0.1\bundle-1.0.1`. Copy what's inside to `C:\hms` and run the installer from there.

> The installer creates `.env`, which holds the keys that decrypt patient data. Tell the hospital to back up `.env` and keep it private.

4 Issue the licence

Licences are created only in the **SaaS console**, by two people, each confirming with a **2FA code** from their authenticator app.

Where the console runs. The SaaS console runs on the SaaS server, which holds the private signing key. Until that server is live, run the console on your PC with the key, in two Command Prompt windows kept open:

```
REM Window 1 - backend with the signing key
cd /d E:\Aniket\next\Hospital\Hospital\Backend
venv\Scripts\python.exe manage.py migrate
set LICENSE_SIGNING_KEY_PATH=C:\Users\User\Desktop\key\license_signing_key.pem
venv\Scripts\python.exe manage.py runserver 8000
```

```
REM Window 2 - the console
cd /d E:\Aniket\next\Hospital\Hospital\Frontend
npm run dev
```

Open **http://localhost:3000** and sign in as the SaaS Owner. `migrate` is needed only after pulling new code; it's safe to run every time.

One-time: turn on 2FA. In the console, open **Settings** → **Two-factor authentication**, scan the QR code with Google or Microsoft Authenticator and confirm a code. Without it, licences can't be requested or approved.

Who can do what

Person	Can
SaaS Owner	Request and sign licences, approve others' requests, revoke, manage the licence team
Staff an Owner allowed ("May request")	Request licences — nothing is signed until an Owner approves
Everyone else (including Platform Admins)	Nothing licence-related

An Owner allows someone in On-Premise Licences → **Licence team** → *Allow licences*. Every staff account has a permanent code such as **PNX-0007**.

Step 1 — Request. Tenants & Subscriptions → the hospital → **License**:

Field	What to enter
Deployment ID	**Paste from their REQUEST-LICENCE.txt.** Never leave it blank.
Bind to one machine	Ticked, then **paste the fingerprint from REQUEST-LICENCE.txt**
Expires on, grace period	From the contract (grace usually 14 days)
Max users, max beds, tier	From the contract
Licensed features	Tick the purchased features
2FA code	Current 6-digit code from your authenticator app

Press **Request / Generate License**. If you are a SaaS Owner the licence is signed and downloads at once. Otherwise it says *Sent to a SaaS Owner for approval*, and the Owners get an email.

Step 2 — Approve (SaaS Owner). On-Premise Licences → **Licence requests** → check the terms and that the hospital has paid → **Approve** → 2FA code. The `.lic` file downloads. Owners get an email for every licence

issued, and the register flags any licence with **No paid invoice**.

If a step fails:

Message	Fix
Turn on two-factor authentication	Do the one-time 2FA setup above
Enter the current 6-digit code	The code expired; type the one showing now
Only SaaS Owners, and staff a SaaS Owner has authorised	Sign in as an Owner, or ask an Owner to allow you
No signing key	`set LICENSE_SIGNING_KEY_PATH` wasn't run before `runserver`. Stop it (Ctrl+C) and run both lines again
Server error on the licence pages	`migrate` wasn't run after pulling new code

The licence has the requester's and approver's staff codes signed inside it. The hospital sees them in **Settings** → **License** → **Issued by**. Anyone can check a file (no key needed):

```
REM Command Prompt on your PC
cd /d E:\Aniket\next\Hospital\Hospital\Backend\tools
..\venv\Scripts\python.exe -m license_issuer verify license-city-hospital.lic
```

It must say **Signature OK**, show `Issued by : PNX-.... (approved by PNX-....)`, and its Deployment ID and fingerprint must match `REQUEST-LICENCE.txt`. Then **send the `.lic` file to the hospital.** It is safe to email: it can't be edited and only works on their server.

When a staff member leaves: Licence team → **Block** → 2FA code. Their sessions end immediately on every device, their licence right is removed and their pending requests are cancelled. Search the register for their code to review every licence they issued, and revoke any that shouldn't exist.

5 Activate

Hospital IT saves the licence as `config/license.lic` in the install folder and runs the installer again.

Linux:

```
# Linux server
sudo cp license-city-hospital.lic /opt/hms/config/license.lic
cd /opt/hms
sudo ./install.sh
```

Windows:

```
# Windows server - PowerShell (administrator)
Copy-Item .\license-city-hospital.lic C:\hms\config\license.lic
cd C:\hms
powershell -ExecutionPolicy Bypass -File .\install.ps1
```

The installer finds the licence, prepares the database, asks for the **administrator email and password**, creates the hospital, and starts everything. It ends by printing the address to open, e.g. `http://hms.cityhospital.local/` (or with `:8080` if port 80 was busy).

The hospital signs in, and checks **Settings** → **License** shows the purchased features, users and expiry.

6 Domain and HTTPS

So staff open `https://hms.hospital.com` instead of an IP address.

1. **DNS:** the hospital's IT points the name at the server. For use inside the hospital (recommended), add it to the internal DNS (Windows Server DNS or the router) with the server's LAN IP. For access from outside, add a public DNS record and forward ports 80 and 443 to the server — a VPN is safer than exposing a hospital system to the internet.
2. **Run the domain script** in the install folder:

```
# Linux server
sudo ./configure-domain.sh
```

```
# Windows server - PowerShell (administrator)
powershell -ExecutionPolicy Bypass -File .\configure-domain.ps1
```

It asks for the domain, then how to do HTTPS:

Choice	When to use it
1. Hospital's own certificate	The hospital has a certificate (e.g. `*.hospital.com`). Give the PEM certificate and key files; they are checked before use.
2. Let's Encrypt	Free, renews itself automatically. The server must be reachable from the internet on port 80.
3. No certificate	Plain HTTP, internal network only, until a certificate is available.

It updates the settings and restarts. Use ports 80 and 443 for a domain: if the installer moved to 8080 because port 80 was busy, free port 80 first (see RUNBOOK section 8).

7 Backups

Every night at 02:00 the system backs up the whole database and uploaded documents into the `backups` folder, keeps 30 days, and writes the result to `backups/LAST-BACKUP.txt`.

Backups must leave the server, or a disk failure loses them too. Hospital IT sets `BACKUP_DIR` in `.env` to a NAS or external drive and restarts the backup service:

```
# .env in the install folder (one of these)
BACKUP_DIR=/mnt/nas/hms-backups
BACKUP_DIR=D:/hms-backups
```

```
# Linux or Windows server, in the install folder
docker compose up -d backup
docker compose exec backup sh /backup.sh now
```

The second line makes a backup immediately, to confirm it works. Then:

- **Weekly:** open `LAST-BACKUP.txt` and check it says `OK`.
- **Once after installing:** practise a restore (RUNBOOK section 9).
- **Keep a copy of `.env` separately and safely:** backups can't be read without its keys.

8 Updates and renewals

New version of the software:

1. You build a new bundle with a higher version (Part 1) and deliver it (Part 2).
2. Hospital IT unpacks it into a **new, separate folder** and runs the upgrade from there:

```
# Linux server (from the new bundle folder)
sudo ./upgrade.sh /opt/hms
```

```
# Windows server - PowerShell (administrator), from the new bundle folder
powershell -ExecutionPolicy Bypass -File .\upgrade.ps1 -InstallDir C:\hms
```

The upgrade **backs up the database first**, then updates and restarts. The licence, settings and all data are kept.

Licence renewal (yearly, or when they buy more users or features): issue a new licence for the same hospital as in Part 4 (request, then Owner approval) with the new expiry, users or features, and the same deployment ID and fingerprint. Ask the hospital for a usage report first (Settings → License → **Download usage report**) and upload it in On-Premise Licences: it shows their real number of users. If the report names a licence you never issued, the upload is refused and the Owners get a security alert.

Send the new file. The hospital admin uploads it in **Settings** → **License** — it applies immediately, no restart.

Full reference: `deploy/RUNBOOK.md` (troubleshooting, HTTPS, rollback). **Practice run:** `docs/TRIAL_INSTALL_WINDOWS.pdf`.